

Section 2.8 Derivatives as Functions

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Ex ① Find the derivative of $f(x) = x^3 - 2$

$$\lim_{x \rightarrow a} \frac{x^3 - 2 - [a^3 - 2]}{x - a}$$

$$\lim_{x \rightarrow a} \frac{x^3 - a^3}{x - a}$$

$$\lim_{x \rightarrow a} \frac{(\cancel{x-a})(x^2 + a \cdot x + a^2)}{\cancel{x-a}}$$

$$= a^2 + a \cdot a + a^2$$

$$= 3a^2 \quad \text{or} \quad \boxed{3x^2} \quad \checkmark$$

Ex ② Find derivative of $f(x) = \frac{1}{x}$

Notation $f'(x)$ means derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\overset{x(x+h)}{x+h} - \frac{1}{x} \overset{x(x+h)}{x}}{h \cdot x(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x} - \cancel{x} - h}{h \cdot x(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-\cancel{h} 1}{\cancel{h} \cdot x(x+h)}$$

$$= \boxed{\frac{-1}{x^2}}$$

undefined when $x=0$.

Notation for derivative

$$f'(x)$$

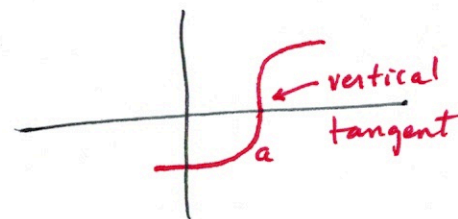
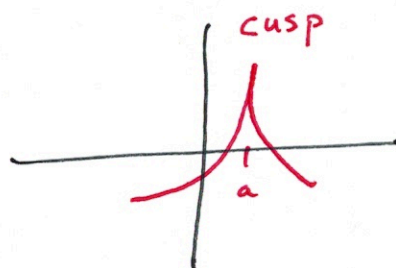
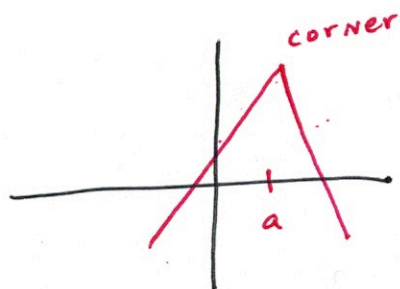
$$y'$$

$$\frac{d(\ddot{ })}{dx}$$

$$\frac{dy}{dx}$$

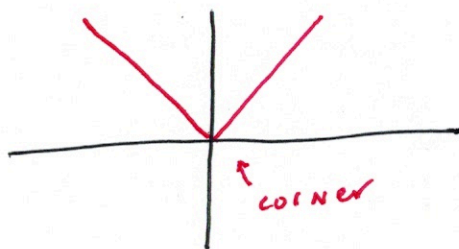
Does the derivative Always exist?

The function must be continuous.



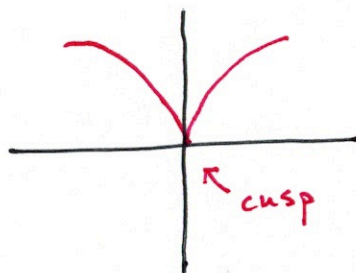
$f'(a)$ does NOT exist.

Ex ③ $y = |x|$ where is y' undefined?



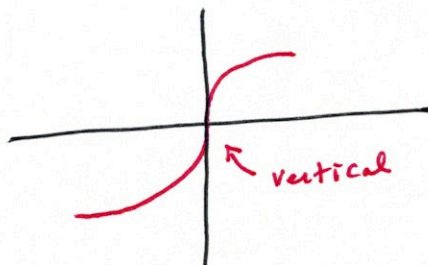
$y'(0)$ DNE.

④ $y = x^{2/3}$ where is y' undefined?



$y'(0)$ DNE

⑤ $y = \sqrt[3]{x} \rightarrow y = x^{1/3} \rightarrow y' = \frac{1}{3} x^{-2/3} \rightarrow y' = \frac{1}{3x^{2/3}}$



$y'(0)$ DNE